ENGR 225: Mechanics 1

Homework Coversheet

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DATE: _____

ASSIGNMENT: Homework 7

Level of involvement of AI, LLMs, or related tools in the completion of this assignment:

☐ None

■ Minor

☐ Major

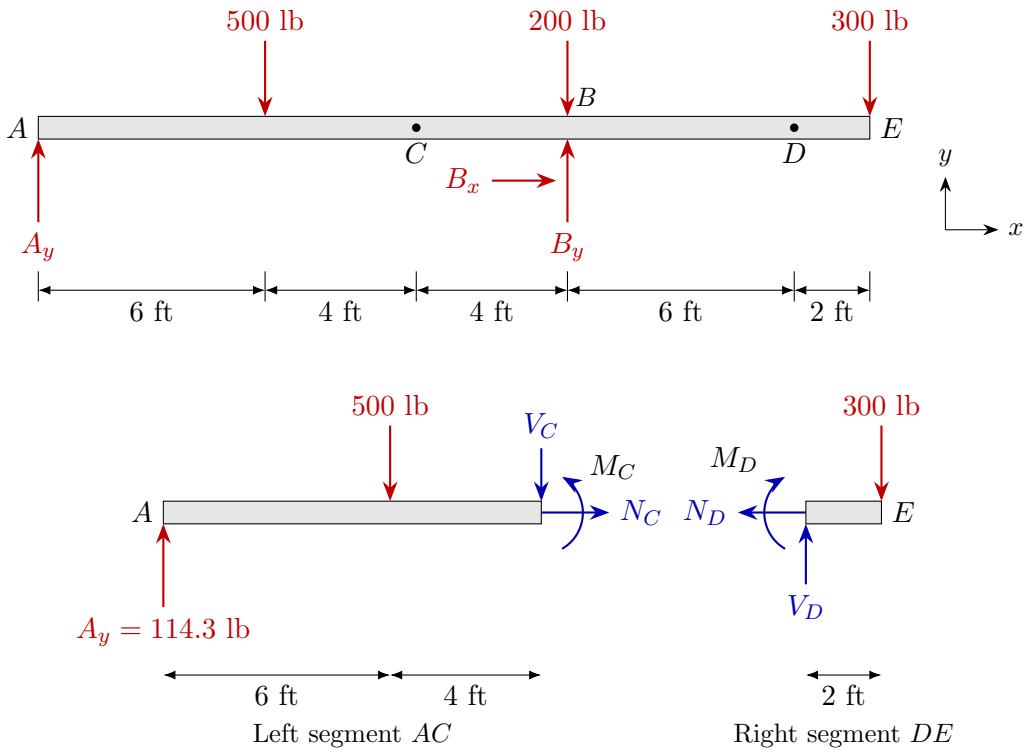
☐ Extensive

Mark the box that applies (e.g., replace ☐ with ■, or type an X beside it, before compiling/printing). Your disclosure does not affect your grade – it keeps you accountable for the work you are doing.

Problem 1

Given / Find: Beam AE with a roller at A and a pin at B. Measured from A: a 500 lb load at 6 ft, point C at 10 ft, B at 14 ft (with a 200 lb load applied at B), point D at 20 ft, and a 300 lb load at end E (22 ft). Find the internal shear V and moment M at C and at D.

Sketch:



Solution:

Reactions ($B_x = 0$):

$$\sum M_B = -A_y(14\,\text{ft}) + (500\,\text{lb})(8\,\text{ft}) - (300\,\text{lb})(8\,\text{ft}) = 0$$

$$A_y = \frac{4000 - 2400}{14}\,\text{lb} = 114.3\,\text{lb}$$

$$\sum F_y = A_y + B_y - 500\,\text{lb} - 200\,\text{lb} - 300\,\text{lb} = 0$$

$$B_y = 1000\,\text{lb} - 114.3\,\text{lb} = 885.7\,\text{lb}$$

Section C, left segment AC:

$$\sum F_y = 114.3\,\text{lb} - 500\,\text{lb} - V_C = 0 \;\Rightarrow\; V_C = -385.7\,\text{lb}$$

$$\sum M_C = M_C - (114.3\,\text{lb})(10\,\text{ft}) + (500\,\text{lb})(4\,\text{ft}) = 0 \;\Rightarrow\; M_C = 1143 - 2000 = -857.1\,\text{lb ft}$$

Section D, right segment DE:

$$\sum F_y = V_D - 300\,\text{lb} = 0 \;\Rightarrow\; V_D = 300\,\text{lb}$$
$$\sum M_D = -M_D - (300\,\text{lb})(2\,\text{ft}) = 0 \;\Rightarrow\; M_D = -600\,\text{lb ft}$$

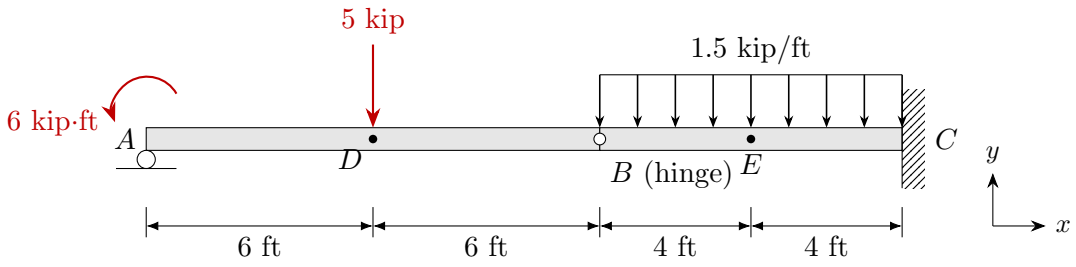
Answer

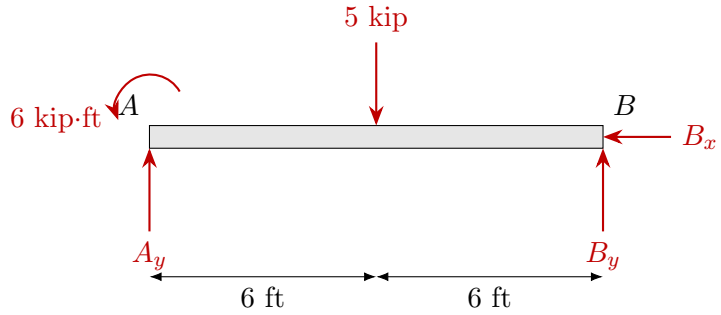
$V_C = -386\,\text{lb}, \quad M_C = -857\,\text{lb ft}, \qquad V_D = 300\,\text{lb}, \quad M_D = -600\,\text{lb ft}$

Problem 2

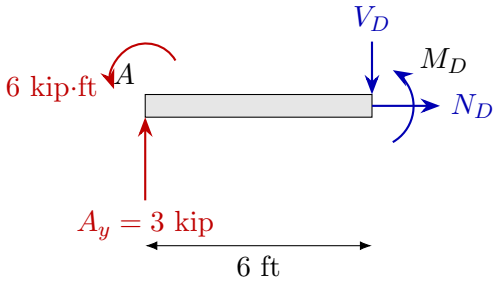
Given / Find: Compound beam: roller at A, internal hinge (pin) at B, fixed support at C. A 6 kip·ft counterclockwise couple acts at A, a 5 kip load acts 6 ft from A (just right of D), B is 12 ft from A, and a uniform 1.5 kip/ft load covers BC (8 ft) with E at its midpoint. Find N, V, and M at sections through D and E.

Sketch:

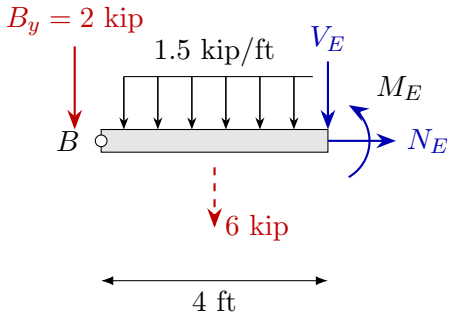




FBD of segment AB ($M_B = 0$ at the hinge)



Left segment AD (5 kip just right of cut)



Segment BE

Solution:

Segment AB (hinge: $M_B = 0$):

$$\sum M_B = -A_y(12\,\text{ft}) + (5\,\text{kip})(6\,\text{ft}) + 6\,\text{kip ft} = 0 \;\Rightarrow\; A_y = \frac{36}{12} = 3.00\,\text{kip}$$

$$\sum F_y = 3\,\text{kip} - 5\,\text{kip} + B_y = 0 \;\Rightarrow\; B_y = 2.00\,\text{kip (on } AB)$$

$$\sum F_x = B_x = 0$$

Hinge force on BC: 2.00 kip $\downarrow$.

Section D, left segment AD (5 kip just right of cut):

$$\begin{aligned}\textstyle\sum F_x &= N_D = 0\\ \textstyle\sum F_y &= 3\,\text{kip} - V_D = 0 \;\Rightarrow\; V_D = 3.00\,\text{kip}\\ \textstyle\sum M_D &= M_D - (3\,\text{kip})(6\,\text{ft}) + 6\,\text{kip ft} = 0 \;\Rightarrow\; M_D = 12.0\,\text{kip ft}\end{aligned}$$

Section E, segment BE (distributed resultant $1.5(4) = 6$ kip, 2 ft from E):

$$\begin{aligned}\textstyle\sum F_x &= N_E = 0\\ \textstyle\sum F_y &= -2\,\text{kip} - 6\,\text{kip} - V_E = 0 \;\Rightarrow\; V_E = -8.00\,\text{kip}\\ \textstyle\sum M_E &= M_E + (2\,\text{kip})(4\,\text{ft}) + (6\,\text{kip})(2\,\text{ft}) = 0 \;\Rightarrow\; M_E = -20.0\,\text{kip ft}\end{aligned}$$

Answer

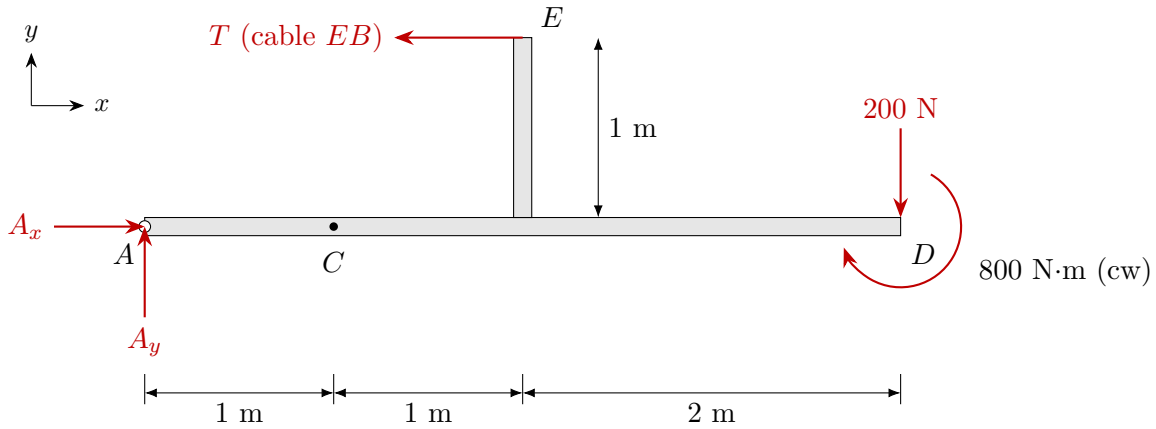
$N_D = 0, \quad V_D = 3.00\,\text{kip}, \quad M_D = 12.0\,\text{kip ft}$

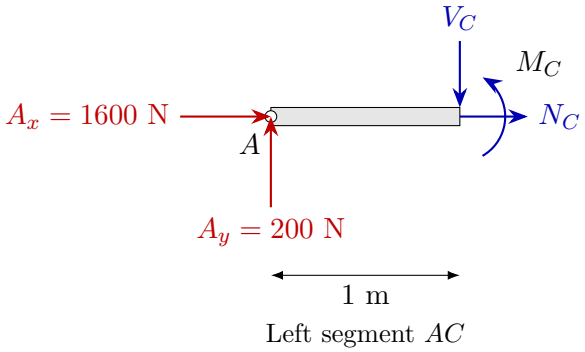
$N_E = 0, \quad V_E = -8.00\,\text{kip}, \quad M_E = -20.0\,\text{kip ft}$

Problem 3

Given / Find: Horizontal member AD pinned to the wall at A. A vertical post rises 1 m from the member at 2 m from A to point E, and a horizontal cable EB ties E back to the wall. At D (4 m from A) a 200 N downward load and an 800 N·m clockwise couple act. Point C is 1 m from A. Find the internal normal force, shear, and moment at C.

Sketch:





Left segment AC

Solution:

Whole frame (T horizontal at E, 1 m above AD):

$$\sum M_A = T(1\,\text{m}) - (200\,\text{N})(4\,\text{m}) - 800\,\text{N m} = 0 \;\Rightarrow\; T = 1600\,\text{N}$$
$$\sum F_x = A_x - 1600\,\text{N} = 0 \;\Rightarrow\; A_x = 1600\,\text{N}$$
$$\sum F_y = A_y - 200\,\text{N} = 0 \;\Rightarrow\; A_y = 200\,\text{N}$$

Section C, left segment AC:

$$\sum F_x = 1600\,\text{N} + N_C = 0 \;\Rightarrow\; N_C = -1600\,\text{N (compression)}$$
$$\sum F_y = 200\,\text{N} - V_C = 0 \;\Rightarrow\; V_C = 200\,\text{N}$$
$$\sum M_C = M_C - (200\,\text{N})(1\,\text{m}) = 0 \;\Rightarrow\; M_C = 200\,\text{N m}$$

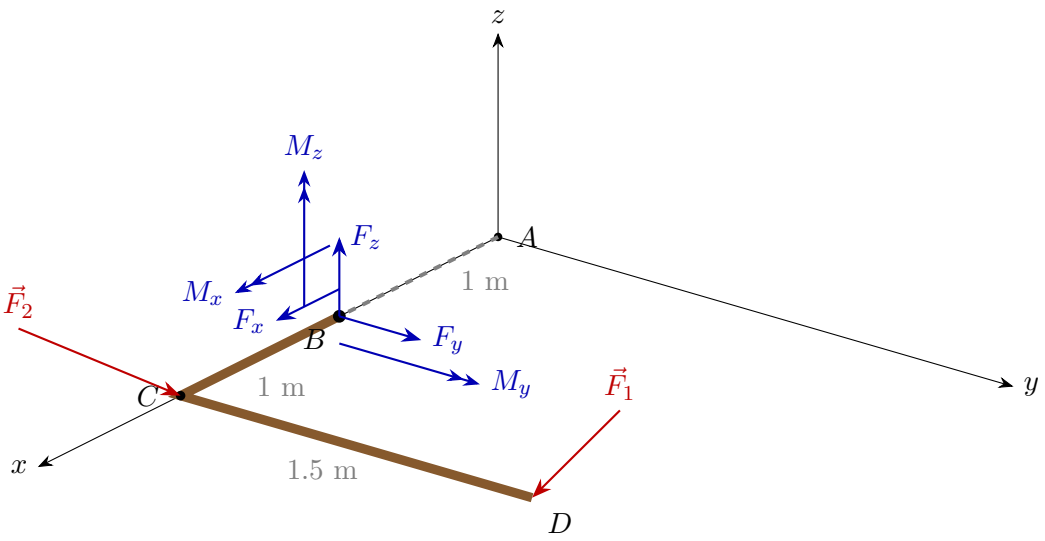
Answer

$N_C = -1.60\,\text{kN}$ (compression), $V_C = 200\,\text{N}$, $M_C = 200\,\text{N m}$

Problem 4

Given / Find: L-shaped pipe fixed at A. Segment A-to-elbow runs 2 m along x, with B at its midpoint (1 m from A). From the elbow, a 1.5 m segment runs in the $+y$ direction. $\vec{F}_2 = (100\hat{\imath} + 500\hat{\jmath})$ N acts at the elbow (labeled C here) and $\vec{F}_1 = (100\hat{\imath} - 200\hat{\jmath} - 300\hat{k})$ N acts at the free end (labeled D here). Pipe weight is neglected. Find the x, y, z components of internal force and moment at a section through B.

Sketch:



FBD of segment BCD (C = elbow, D = free end). Internal loads at B drawn in the $+x$, $+y$, $+z$ directions (single arrow = force, double arrow = moment); a negative answer means the actual load points the other way.

Solution:

Outboard segment BCD, position vectors from B:

$$\vec{r}_{BC} = (1\hat{\imath})\,\text{m}, \qquad \vec{r}_{BD} = (1\hat{\imath} + 1.5\hat{\jmath})\,\text{m}$$

Force equilibrium:

$$\begin{aligned}\sum \vec{F} &= \vec{F}_B + \vec{F}_1 + \vec{F}_2 = 0\\ \vec{F}_B &= -\big[(100+100)\hat{\imath} + (-200+500)\hat{\jmath} + (-300)\hat{k}\big]\,\text{N}\\ &= (-200\hat{\imath} - 300\hat{\jmath} + 300\hat{k})\,\text{N}\end{aligned}$$

Moment equilibrium about B:

$$\sum \vec{M}_B = \vec{M}_B + \vec{r}_{BD} \times \vec{F}_1 + \vec{r}_{BC} \times \vec{F}_2 = 0$$

$$\vec{r}_{BD} \times \vec{F}_1 = \begin{vmatrix} \hat{\imath} & \hat{\jmath} & \hat{k} \\ 1 & 1.5 & 0 \\ 100 & -200 & -300 \end{vmatrix} = \big[(1.5)(-300) - (0)(-200)\big]\hat{\imath} - \big[(1)(-300) - (0)(100)\big]\hat{\jmath}$$
$$+ \big[(1)(-200) - (1.5)(100)\big]\hat{k}$$
$$= (-450\hat{\imath} + 300\hat{\jmath} - 350\hat{k})\,\mathrm{N\,m}$$

$$\vec{r}_{BC} \times \vec{F}_2 = \begin{vmatrix} \hat{\imath} & \hat{\jmath} & \hat{k} \\ 1 & 0 & 0 \\ 100 & 500 & 0 \end{vmatrix} = (0)\hat{\imath} - (0)\hat{\jmath} + \big[(1)(500) - (0)(100)\big]\hat{k}$$
$$= (500\hat{k})\,\mathrm{N\,m}$$

$$\vec{M}_B = -\big[-450\hat{\imath} + 300\hat{\jmath} + (-350 + 500)\hat{k}\big]\,\mathrm{N\,m}$$
$$= (450\hat{\imath} - 300\hat{\jmath} - 150\hat{k})\,\mathrm{N\,m}$$

Answer

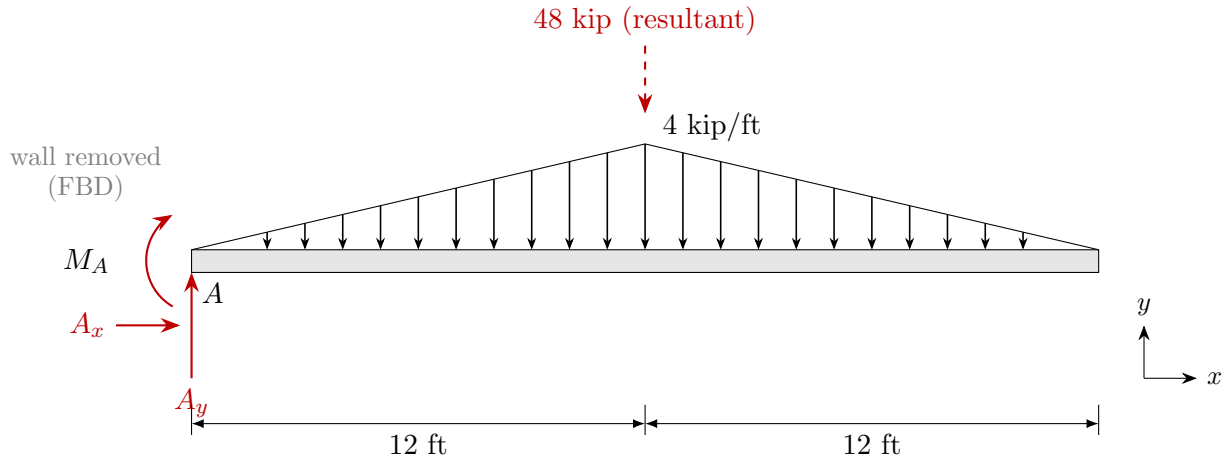
$F_x = -200\,\mathrm{N}, \quad F_y = -300\,\mathrm{N}, \quad F_z = 300\,\mathrm{N}$

$M_x = 450\,\mathrm{N\,m}, \quad M_y = -300\,\mathrm{N\,m}, \quad M_z = -150\,\mathrm{N\,m}$

Problem 5

Given / Find: 24 ft cantilever fixed at A ($x = 0$) carrying a triangular distributed load that rises from 0 at A to 4 kip/ft at $x = 12$ ft and falls back to 0 at the free end ($x = 24$ ft). Find $V(x)$ and $M(x)$ for both regions and draw the shear and moment diagrams.

Sketch:



Solution:

Reactions: $F_R = \frac{1}{2}(24)(4) = 48\,\text{kip}$ at $x = 12$ ft; $A_x = 0$.

$$\sum F_y = A_y - 48\,\text{kip} = 0 \;\Rightarrow\; A_y = 48.0\,\text{kip}$$
$$\sum M_A = M_A - (48\,\text{kip})(12\,\text{ft}) = 0 \;\Rightarrow\; M_A = 576\,\text{kip ft}$$

So $M(0) = -576\,\text{kip ft}$.

$w(x) = \frac{x}{3}$ $(0 \le x \le 12)$, $\quad w(x) = \frac{24-x}{3}$ $(12 \le x \le 24)$ kip/ft.

Region 1, $0 \le x \le 12$ *(left segment):* load $\frac{x^2}{6}$ kip at $\frac{x}{3}$ from cut.

$$\sum F_y = 48 - \frac{x^2}{6} - V = 0 \;\Rightarrow\; V(x) = -\frac{1}{6}x^2 + 48 \quad \text{(kip)}$$
$$\sum M_{\text{cut}} = M + 576 - 48x + \frac{x^2}{6}\cdot\frac{x}{3} = 0$$
$$M(x) = -\frac{1}{18}x^3 + 48x - 576 \quad \text{(kip·ft)}$$

Region 2, $12 \le x \le 24$ *(right segment):* load $\frac{(24-x)^2}{6}$ kip at $\frac{24-x}{3}$ from cut.

$$\sum F_y = V - \frac{(24-x)^2}{6} = 0 \;\Rightarrow\; V(x) = \frac{1}{6}x^2 - 8x + 96 \quad \text{(kip)}$$

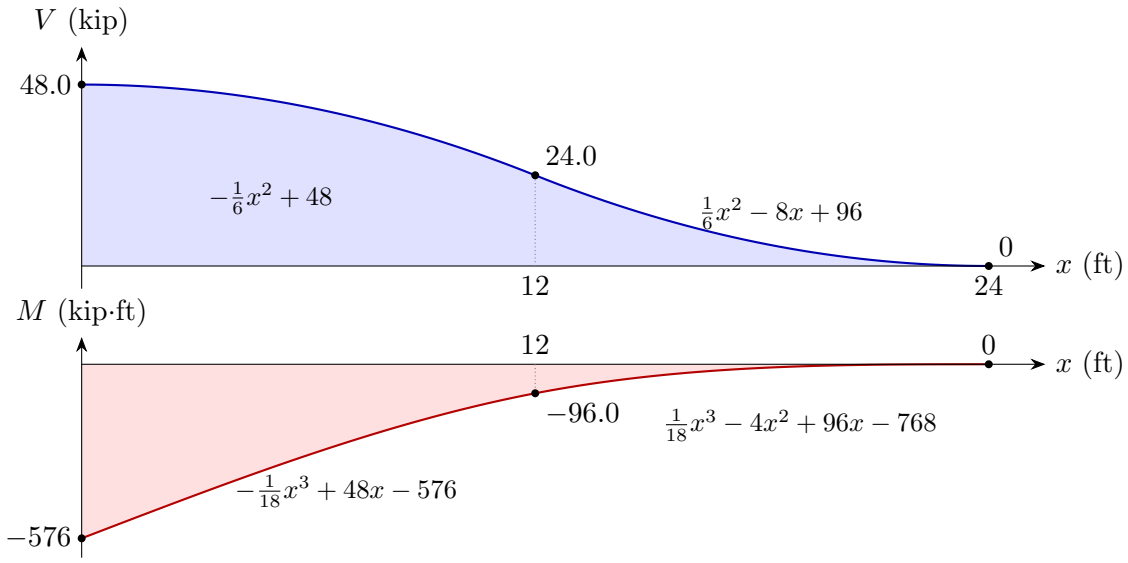
$$\sum M_{\text{cut}} = -M - \frac{(24-x)^2}{6} \cdot \frac{24-x}{3} = 0 \;\Rightarrow\; M(x) = -\frac{(24-x)^3}{18}$$
$$M(x) = \frac{1}{18}x^3 - 4x^2 + 96x - 768 \quad \text{(kip·ft)}$$

Critical values:

$V(0) = 48.0\,\text{kip},$	$V(12) = 24.0\,\text{kip},$	$V(24) = 0$
$M(0) = -576\,\text{kip ft},$	$M(12) = -96.0\,\text{kip ft},$	$M(24) = 0$

$dV/dx = -w \le 0$ and $dM/dx = V \ge 0$ on $0 < x < 24$: no interior zero crossings.

Shear and moment diagrams:



Answer

$0 \le x \le 12$ ft: $V(x) = -\frac{1}{6}x^2 + 48$ kip, $M(x) = -\frac{1}{18}x^3 + 48x - 576$ kip·ft

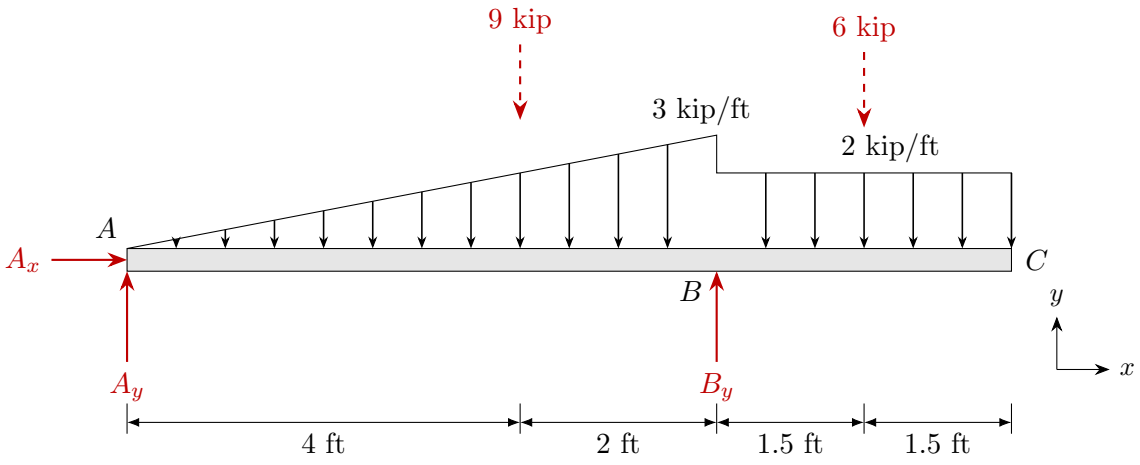
$12 \le x \le 24$ ft: $V(x) = \frac{1}{6}x^2 - 8x + 96$ kip, $M(x) = \frac{1}{18}x^3 - 4x^2 + 96x - 768$ kip·ft

$V_{\max} = 48.0$ kip and $M_{\min} = -576$ kip·ft, both at $x = 0$ (wall); $V(12) = 24.0$ kip, $M(12) = -96.0$ kip·ft; $V = M = 0$ at $x = 24$ ft.

Problem 6

Given / Find: Beam with a pin at A ($x = 0$), a roller at B ($x = 6$ ft), and an overhang to C ($x = 9$ ft). A triangular load rises from 0 at A to 3 kip/ft at B; a uniform 2 kip/ft load covers BC. Find $V(x)$ and $M(x)$ and draw the shear and moment diagrams.

Sketch:



Solution:

Reactions: triangle $\frac{1}{2}(6)(3) = 9\,\text{kip}$ at 4 ft; uniform $2(3) = 6\,\text{kip}$ at 7.5 ft; $A_x = 0$.

$$\sum M_A = B_y(6\,\text{ft}) - (9\,\text{kip})(4\,\text{ft}) - (6\,\text{kip})(7.5\,\text{ft}) = 0 \;\Rightarrow\; B_y = \frac{36 + 45}{6} = 13.5\,\text{kip}$$

$$\sum F_y = A_y + 13.5 - 9 - 6 = 0 \;\Rightarrow\; A_y = 1.50\,\text{kip}$$

Region 1, $0 \le x < 6$ *(left segment):* $w = \frac{x}{2}$, load $\frac{x^2}{4}$ kip at $\frac{x}{3}$ from cut.

$$\sum F_y = 1.5 - \frac{x^2}{4} - V = 0 \;\Rightarrow\; V(x) = -\frac{1}{4}x^2 + 1.5 \quad \text{(kip)}$$

$$\sum M_{\text{cut}} = M - 1.5x + \frac{x^2}{4} \cdot \frac{x}{3} = 0 \;\Rightarrow\; M(x) = -\frac{1}{12}x^3 + 1.5x \quad \text{(kip·ft)}$$

Region 2, $6 < x \le 9$ *(right segment):* load $2(9 - x)$ kip at $\frac{9-x}{2}$ from cut.

$$\sum F_y = V - 2(9 - x) = 0 \;\Rightarrow\; V(x) = -2x + 18 \quad \text{(kip)}$$

$$\sum M_{\text{cut}} = -M - 2(9 - x) \cdot \frac{9 - x}{2} = 0$$

$$M(x) = -(9 - x)^2 = -x^2 + 18x - 81 \quad \text{(kip·ft)}$$

Critical values:

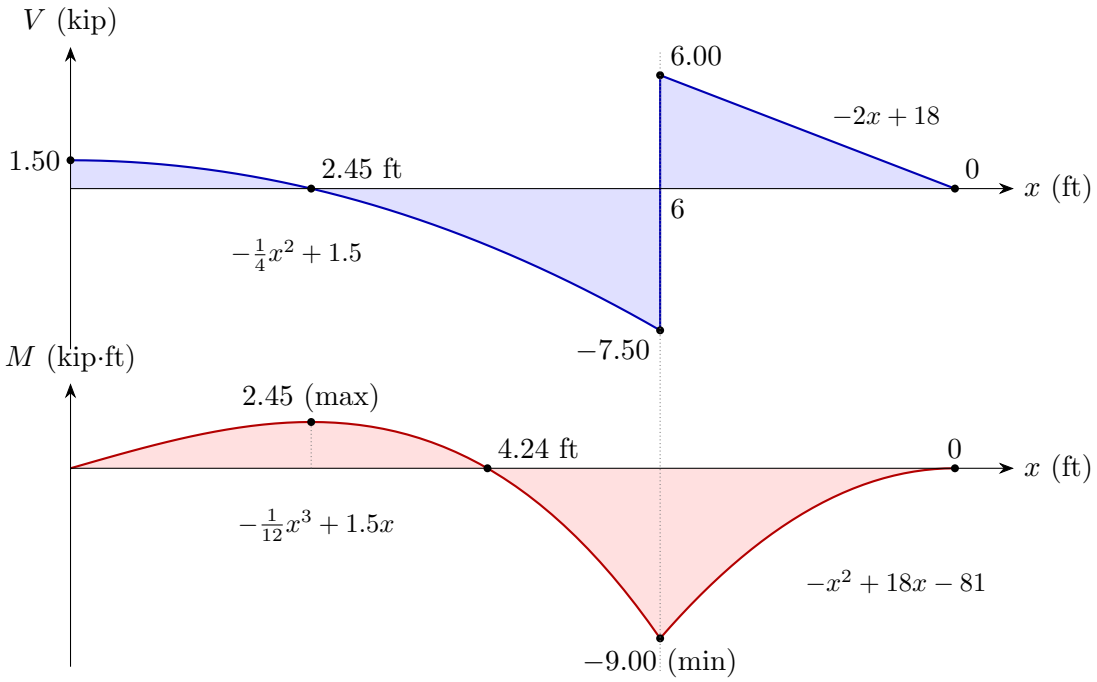
$$V = 0: \ -\tfrac{1}{4}x^2 + 1.5 = 0 \Rightarrow x = 2.45\,\text{ft}, \quad M_{\max} = -\tfrac{1}{12}(2.449)^3 + 1.5(2.449) = 2.45\,\text{kip ft}$$

$$M = 0: \ -\tfrac{1}{12}x^3 + 1.5x = 0 \Rightarrow x = \sqrt{18} = 4.24\,\text{ft}$$

$$x = 6: \ V(6^-) = -7.50\,\text{kip}, \ V(6^+) = 6.00\,\text{kip} \ (\Delta V = B_y), \quad M(6) = -9.00\,\text{kip ft}$$

$$x = 9: \ V = 0, \quad M = 0$$

Shear and moment diagrams:



Answer

$0 \le x < 6$ ft: $V(x) = -\tfrac{1}{4}x^2 + 1.5$ kip, $M(x) = -\tfrac{1}{12}x^3 + 1.5x$ kip·ft

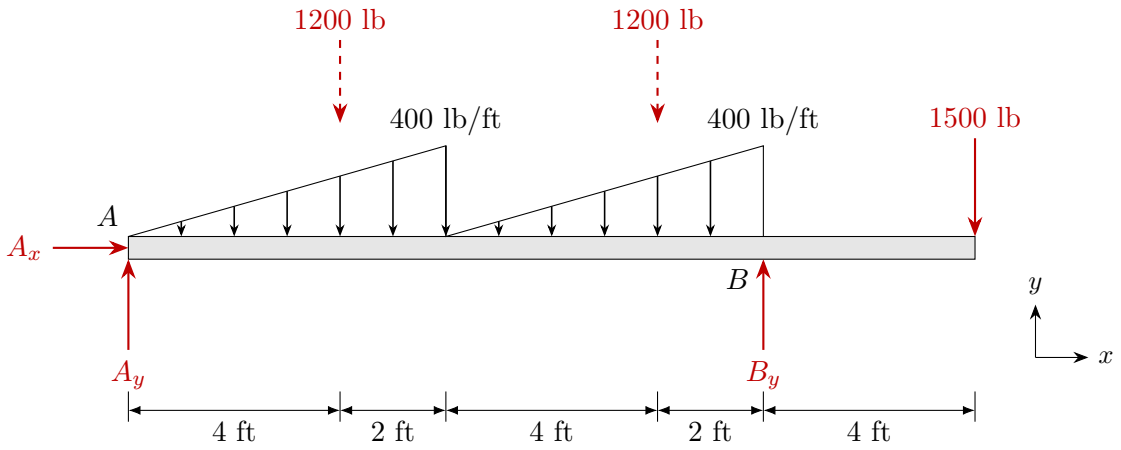
$6 < x \le 9$ ft: $V(x) = -2x + 18$ kip, $M(x) = -x^2 + 18x - 81$ kip·ft

$V = 0$ and $M_{\max} = 2.45$ kip·ft at $x = 2.45$ ft; $M = 0$ at $x = 4.24$ ft; at B, V jumps from -7.50 to 6.00 kip and $M_{\min} = -9.00$ kip·ft.

Problem 7

Given / Find: Beam with a pin at A ($x = 0$), a roller at B ($x = 12$ ft), and a 1500 lb load at the free end ($x = 16$ ft). Two identical triangular loads act: each rises from 0 to 400 lb/ft over 6 ft (from $x = 0$ to 6 and from $x = 6$ to 12). Find $V(x)$ and $M(x)$ and draw the shear and moment diagrams.

Sketch:



Solution:

Reactions: each triangle $\frac{1}{2}(6)(400) = 1200\,\text{lb}$, at $x = 4$ and $x = 10$ ft; $A_x = 0$.

$$\sum M_A = B_y(12) - 1200(4) - 1200(10) - 1500(16) = 0$$

$$B_y = \frac{4800 + 12\,000 + 24\,000}{12}\,\text{lb} = 3400\,\text{lb}$$

$$\sum F_y = A_y + 3400 - 1200 - 1200 - 1500 = 0 \;\Rightarrow\; A_y = 500\,\text{lb}$$

Region 1, $0 \le x \le 6$ *(left segment):* $w = 66.7x$, load $33.3x^2$ lb at $\frac{x}{3}$ from cut.

$$\sum F_y = 500 - 33.3x^2 - V = 0 \;\Rightarrow\; V(x) = -33.3x^2 + 500 \quad \text{(lb)}$$

$$\sum M_{\text{cut}} = M - 500x + 33.3x^2 \cdot \frac{x}{3} = 0 \;\Rightarrow\; M(x) = -11.1x^3 + 500x \quad \text{(lb·ft)}$$

Region 2, $6 \le x < 12$ *(left segment):* first triangle 1200 lb at $x = 4$; second $33.3(x-6)^2$ lb at $\frac{x-6}{3}$ from cut.

$$\sum F_y = 500 - 1200 - 33.3(x-6)^2 - V = 0$$

$$V(x) = -700 - 33.3(x^2 - 12x + 36)$$

$$= -33.3x^2 + 400x - 1900 \quad \text{(lb)}$$

$$\sum M_{\text{cut}} = M - 500x + 1200(x-4) + 33.3(x-6)^2 \cdot \frac{x-6}{3} = 0$$

$$M(x) = -700x + 4800 - 11.1(x^3 - 18x^2 + 108x - 216)$$

$$= -11.1x^3 + 200x^2 - 1900x + 7200 \quad \text{(lb·ft)}$$

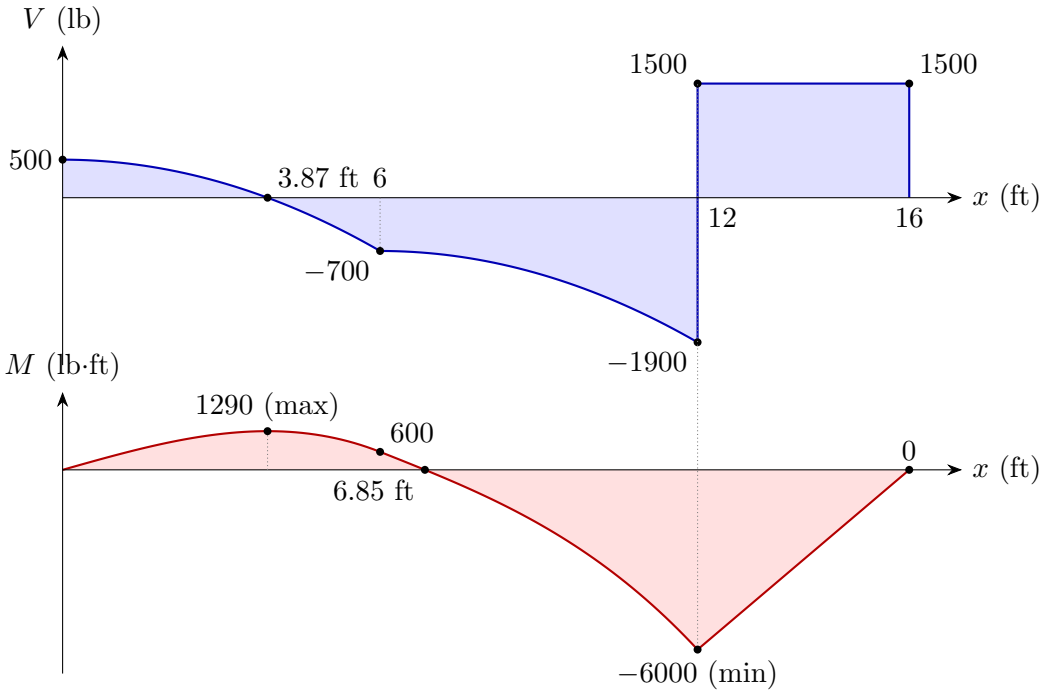
Region 3, $12 < x \le 16$ (right segment):

$$\sum F_y = V - 1500 = 0 \;\Rightarrow\; V(x) = 1500\,\text{lb}$$
$$\sum M_{\text{cut}} = -M - 1500(16 - x) = 0 \;\Rightarrow\; M(x) = 1500x - 24\,000 \quad (\text{lb·ft})$$

Critical values:

$$V = 0: \; 500 - 33.3x^2 = 0 \Rightarrow x = 3.87\,\text{ft}$$
$$M_{\max} = 500(3.873) - 11.1(3.873)^3 = 1290\,\text{lb ft}$$
$$x = 6: \; V = -700\,\text{lb}, \quad M = 600\,\text{lb ft}$$
$$M = 0 \; (u = x - 6): \; u^3 + 63u - 54 = 0 \Rightarrow u = 0.847, \quad x = 6.85\,\text{ft}$$
$$x = 12: \; V(12^-) = -1900\,\text{lb}, \; V(12^+) = 1500\,\text{lb} \; (\Delta V = B_y), \quad M(12) = -6000\,\text{lb ft}$$
$$x = 16: \; V \to 0, \quad M = 0$$

Shear and moment diagrams:



Answer

$0 \le x \le 6$ ft: $V(x) = -33.3x^2 + 500$ lb, $M(x) = -11.1x^3 + 500x$ lb·ft

$6 \le x < 12$ ft: $V(x) = -33.3x^2 + 400x - 1900$ lb, $M(x) = -11.1x^3 + 200x^2 - 1900x + 7200$ lb·ft

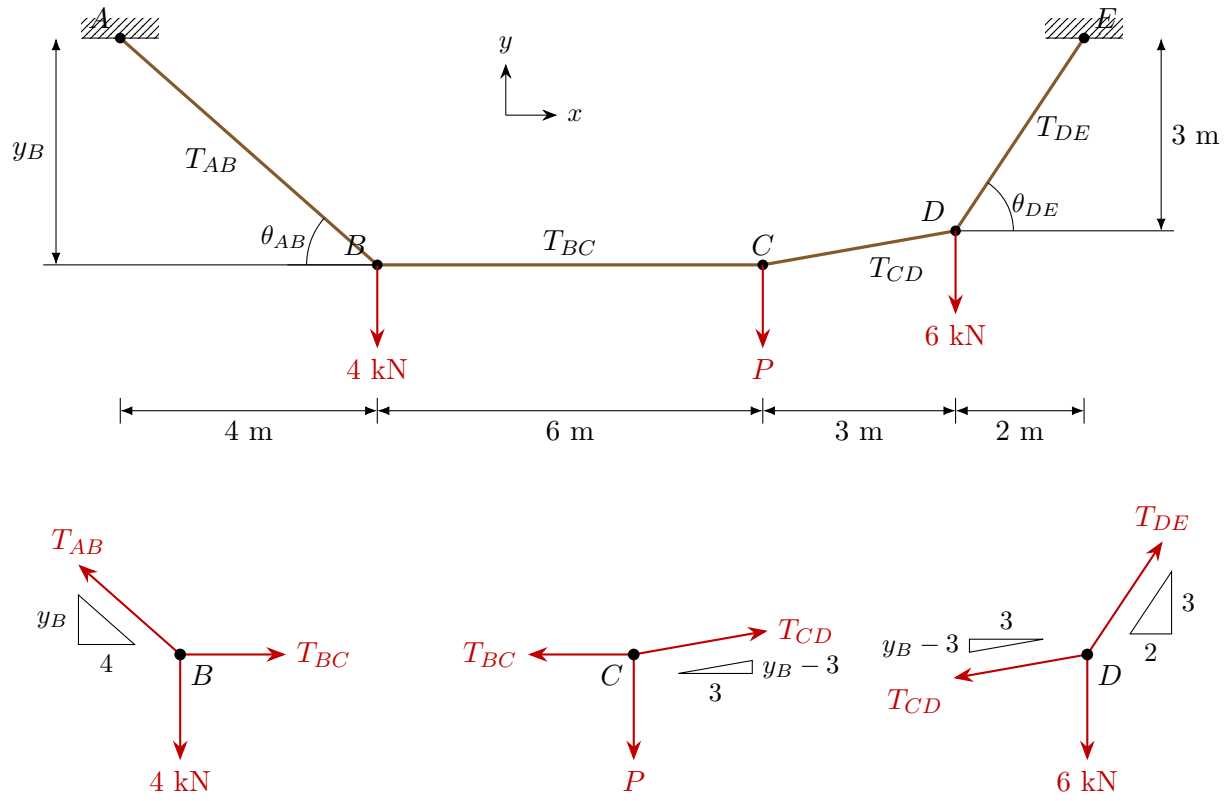
$12 < x \le 16$ ft: $V(x) = 1500$ lb, $M(x) = 1500x - 24\,000$ lb·ft

$M_{\max} = 1290$ lb·ft at $x = 3.87$ ft ($V = 0$); $M = 0$ at $x = 6.85$ ft; $M_{\min} = -6000$ lb·ft at B, where V jumps from -1900 to 1500 lb.

Problem 8

Given / Find: Cable $ABCDE$ hangs between supports A and E at the same elevation. Horizontal spacing: A–B 4 m, B–C 6 m, C–D 3 m, D–E 2 m. Loads: 4 kN at B, P at C, 6 kN at D. D hangs 3 m below E, and segment BC must stay horizontal. Find P, the sag y_B, and the maximum cable tension.

Sketch:



Joint FBDs at B, C, D (tensions pull away from each joint; triangles give segment slopes in m)

Solution:

Geometry (A, E level; B, C at depth y_B; D at 3 m):

$$\tan\theta_{AB} = \frac{y_B}{4}, \qquad \theta_{BC} = 0, \qquad \tan\theta_{CD} = \frac{y_B - 3}{3}, \qquad \tan\theta_{DE} = \frac{3}{2}$$

$\sum F_x = 0$ at each joint $\Rightarrow$ constant horizontal component $T_H = T_{BC}$; vertical component of each segment $= T_H \tan\theta$.

$\sum F_y = 0$ *at each joint:*

$$\text{Joint } B: \quad T_H \frac{y_B}{4} - 4\,\text{kN} = 0 \tag{1}$$

$$\text{Joint } C: \quad T_H \frac{y_B - 3}{3} - P = 0 \tag{2}$$

$$\text{Joint } D: \quad T_H \frac{3}{2} - T_H \frac{y_B - 3}{3} - 6\,\text{kN} = 0 \tag{3}$$

Solve (1) and (3) for T_H, y_B*:* from (1), $T_H = 16/y_B$; into (3):

$$\frac{16}{y_B}\left[\frac{3}{2} - \frac{y_B - 3}{3}\right] = 6 \;\Rightarrow\; \frac{16}{y_B} \cdot \frac{15 - 2y_B}{6} = 6 \;\Rightarrow\; 240 - 32y_B = 36y_B \;\Rightarrow\; y_B = \frac{240}{68} = 3.529\,\text{m}$$

$$T_H = \frac{16\,\text{kN m}}{3.529\,\text{m}} = 4.533\,\text{kN}$$

From (2):

$$P = (4.533\,\text{kN})\frac{3.529 - 3}{3} = (4.533\,\text{kN})(0.1765) = 0.800\,\text{kN}$$

Tensions, $T = T_H\sqrt{1 + \tan^2\theta}$:

$$T_{AB} = 4.533\sqrt{1 + (3.529/4)^2} = 4.533(1.334) = 6.05\,\text{kN}$$
$$T_{BC} = T_H = 4.53\,\text{kN}$$
$$T_{CD} = 4.533\sqrt{1 + (0.529/3)^2} = 4.533(1.015) = 4.60\,\text{kN}$$
$$T_{DE} = 4.533\sqrt{1 + (3/2)^2} = 4.533(1.803) = 8.17\,\text{kN}$$

Answer

$P = 0.800\,\text{kN}, \quad y_B = 3.53\,\text{m}, \quad T_{\max} = T_{DE} = 8.17\,\text{kN}$